

Continuous Casting Machine (CCM) - Standard Operating Procedures

W and S Unit Hydraulic Valve Replacement

- Step 1. Verify the continuous caster strand is fully stopped and the dummy bar is mechanically secured.
- Step 2. Execute Lockout Tagout on the main Hydraulic Power Unit powering the W and S zone.
- Step 3. Isolate the local nitrogen bladder accumulator for the specific pinch roll segment.
- Step 4. Manually bleed residual line pressure via the manifold dump valve until the local analog gauge reads zero bar.
- Step 5. Disconnect the electrical M12 connectors from the valve's solenoids and the LVDT position sensor.
- Step 6. Unbolt the proportional directional valve from the manifold block using a cross pattern sequence to prevent thread binding.
- Step 7. Inspect the manifold mating surface for scoring, and verify all old O rings are removed.
- Step 8. Mount the replacement valve with new O rings, torquing the bolts to the specified limit in a cross pattern.
- Step 9. Reconnect the solenoid and LVDT electrical connectors.
- Step 10. Remove Lockout Tagout, slowly open the isolation block to pressurize the segment, and actuate the manual override to bleed trapped air.
- Step 11. Confirm roll gap calibration and valve stroke response via the SCADA interface.

Dummy Bar Insertion and Mold Sealing (Start-Up)

- Step 1. Inspect the copper mold inner surfaces for wear, scoring, or residue, and verify that the mold oscillator mechanism is free of mechanical play.
- Step 2. Drive the dummy bar from the storage area up through the secondary cooling strand until the dummy bar head seats squarely into the bottom of the mold cavity.
- Step 3. Pack the gaps around the dummy bar head using ceramic fiber rope or asbestos packing sheets to prevent liquid steel leakage during the initial pour.
- Step 4. Place the specified steel or copper cone plug into the dummy bar head cavity to lock the starting head in place.
- Step 5. Pour the required amount of mold casting powder uniformly across the cavity to provide proper lubrication and thermal insulation.
- Step 6. Confirm that primary cooling water headers are pressurized to specification (inlet pressure around 4 bar, outlet around 2 bar) and flow interlocks are green before clearing the area for hot metal.

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Tundish Preheating and Ladle Teeming Preparation

- Step 1. Inspect the tundish refractory lining for cracks, spalling, or excessive wear, and ensure the slide gate or stopper rod mechanism moves freely.
- Step 2. Position the tundish over the preheating station and ignite the burners, ramping the temperature up gradually to prevent thermal shock, targeting a steady state above 1,000 degrees Celsius.
- Step 3. Check that the submerged entry nozzle (SEN) or metering nozzle is securely bolted, aligned, and free of moisture or carbon deposits.
- Step 4. Move the preheated tundish via the tundish car into the casting position directly above the mold centerline using manual alignment rods.
- Step 5. Fill the tundish nozzle bore with the approved neutral filler sand or opening compound to prevent premature freezing when the steel hits.
- Step 6. Verify that emergency oxygen lances and gas regulators on the casting platform are fully charged, leak-free, and set to 2.5 to 3 bar operating pressure.

Emergency Mold Breakout Response

- Step 1. Stop the strand withdrawal drive immediately upon receiving a breakout warning from the breakout prediction system or sighting molten steel dropping below the mold table.
- Step 2. Close the tundish slide gate or drop the stopper rod instantly to cut off the flow of liquid steel into the mold.
- Step 3. Cut off the ladle slide gate to isolate the main steel source from the tundish.
- Step 4. Activate the emergency cooling water dump valves to flood the affected strand zone and freeze the remaining liquid shell safely.
- Step 5. Evacuate the immediate casting pit area and keep all personnel clear of the strand path until thermal contraction ceases.
- Step 6. Inspect the copper mold plates, frame assembly, and containment rolls for thermal distortion and mechanical damage before scheduling a rebuild.